

ADMM: local work and shared capacity

a Commodity-local update

$$x_k \geq 0, \quad Bx_k = b_k$$

$$\min c^T x_k + \frac{\rho}{2} \|x_k - z_k + w_k\|^2$$



b Shared capacity projection

$$z \geq 0, \quad \sum_k z_{ka} \leq u_a$$

Arc-wise Euclidean projection



c Scaled-dual update

$$w \leftarrow w + x - z$$

Fixed, input-derived ρ in selected R2_S



d Independent evaluation

Original-unit conservation, capacity, KKT

Objective and physical back-projection